# MAS8600_assignment

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2026-01-01

## Introduction

Learning Analytics has emerged as a pivotal area of focus within the field of Data Science, with the potential to significantly enhance the understanding of how students engage with educational content and environments. Defined as the "measurement, collection, analysis, and reporting of data about learners and their contexts," Learning Analytics plays a crucial role in optimizing learning experiences and outcomes. By leveraging diverse datasets, Learning Analytics aims to provide valuable insights into the effectiveness of educational methods, inform intervention strategies, and ultimately enhance student performance.

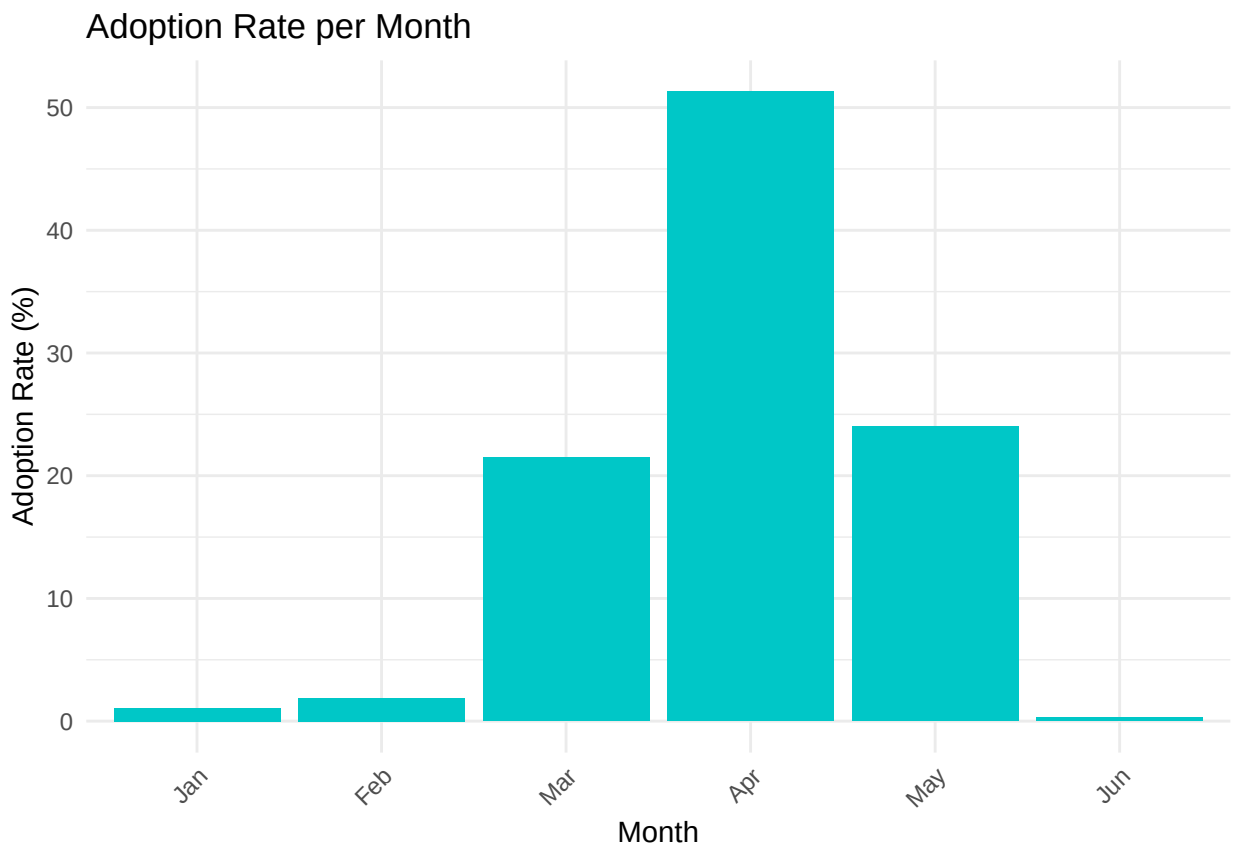
Traditionally, student engagement has been tracked through basic metrics such as attendance, which, while useful, fail to capture the full extent and quality of engagement. Beyond the classroom, various forms of data—ranging from interactions with Virtual Learning Environments (VLEs) and campus facilities to wellbeing referrals—are routinely collected. However, these datasets are often siloed and lack the integration needed to generate comprehensive insights.

This report focuses on a dataset collected from seven runs of the massive open online course (MOOC) titled "Cyber Security: Safety at Home, Online, and in Life," developed by Newcastle University and offered to the public via the FutureLearn platform. The data encompasses learner interactions throughout the course, along with profile-based characteristics, allowing for a deeper understanding of learner behavior and engagement. By aggregating and analyzing this data, this report seeks to identify patterns of engagement, determine factors that influence student success, and explore ways to enhance learning design and student interventions, especially for those at risk of underperformance.

Through the application of Learning Analytics, we aim to not only understand current learner engagement but also optimize future learning experiences, improving both student satisfaction and attainment.

```r
#EDA : Adoption Rate
# Step 3: Create a bar graph of adoption rate per month
ggplot(adoption_pt_enrolments_7,
       aes(x = factor(month_enroled), y = adoption_rate)) +
  geom_bar(stat = "identity", fill = "#00C7C7") +
  labs(title = "Adoption Rate per Month",
       x = "Month",
       y = "Adoption Rate (%)") +
  scale_x_discrete(labels = c(
    "Jan", "Feb", "Mar", "Apr", "May", "Jun",
    "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"
  )) +
```

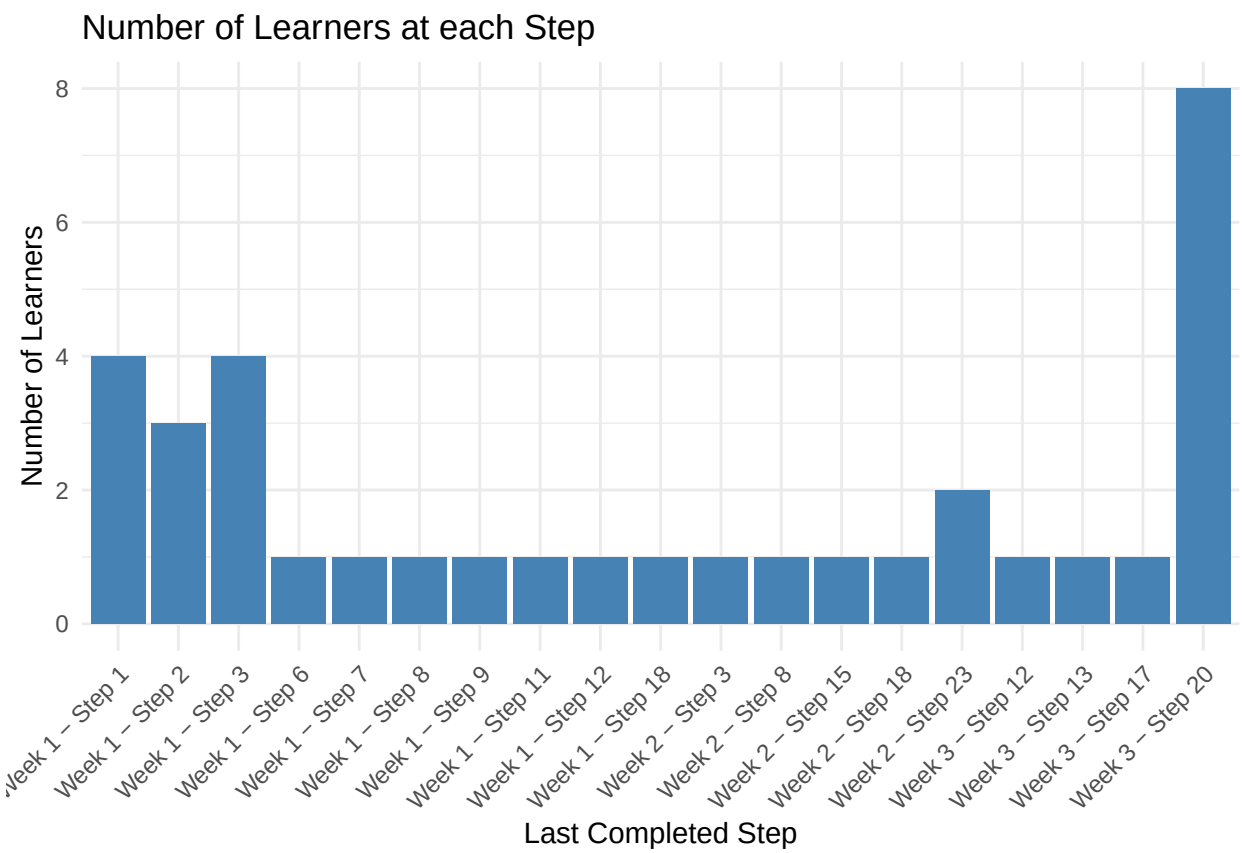
```r
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```




```r
#completion rate
print(paste("Less than 1% completed the course:", total_completion_rate))
```

```
## [1] "Less than 1% completed the course: 1.83603757472246"
```

```r
#Identifying the drop off point
ggplot(drop_off_ordered, aes(x = week_step, y = number_of_learners)) +
  geom_bar(stat = "identity", fill = "steelblue") +
  labs(
    title = "Number of Learners at each Step",
    x = "Last Completed Step",
    y = "Number of Learners"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```




```r
#checking to see if the experience influences when a user leaves
cor(weekly_survey_responses$experience_rating,
    weekly_survey_responses$week_number,
    method = "spearman", use="complete.obs")
```

```
## [1] 0.01525396
```

Positive sign → very slight tendency that higher experience ratings correspond to later leaving weeks Magnitude is very close to 0 → practically no monotonic association Essentially, experience rating and leaving week are independent in your dataset.

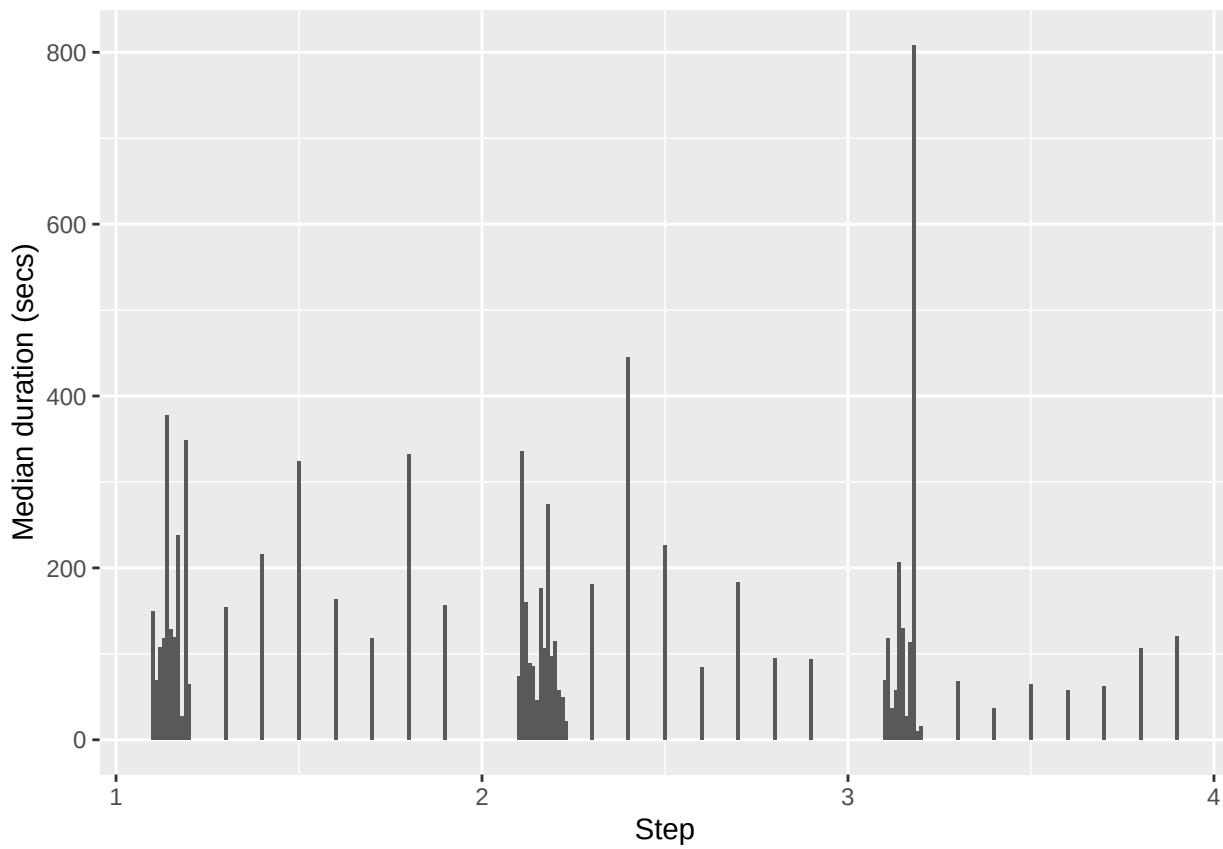
```r
#does the video duration influence last_completed_week number. intereprétation, the users that complete
cor.test(df$video_duration, df$last_completed_week_number, method="spearman")
```

```
## Warning in cor.test.default(df$video_duration, df$last_completed_week_number, :
## Cannot compute exact p-value with ties

##
##  Spearman's rank correlation rho
##
## data:  df$video_duration and df$last_completed_week_number
## S = 0, p-value < 2.2e-16
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
## rho
##   1
```

As video_duration increases, last_completed_week_number always increases without exception. Strength: rho = 1 → extremely strong positive monotonic relationship Direction: positive → longer videos correspond to later leaving weeks Significance: p < 2.2e-16 → statistically significant Caution: ties reduce the precision of the exact p-value, but the conclusion of strong positive correlation is still valid. In practice: rho = 1 with discrete steps often indicates that step number is perfectly aligned with cumulative video duration — which makes sense if video_duration is taken from the step the learner last completed.

```
#how long does it take to complete one step
ggplot(summary_stats, aes(x=step, y=median_duration)) +
  geom_col() +
  labs(y="Median duration (secs)", x="Step")
```

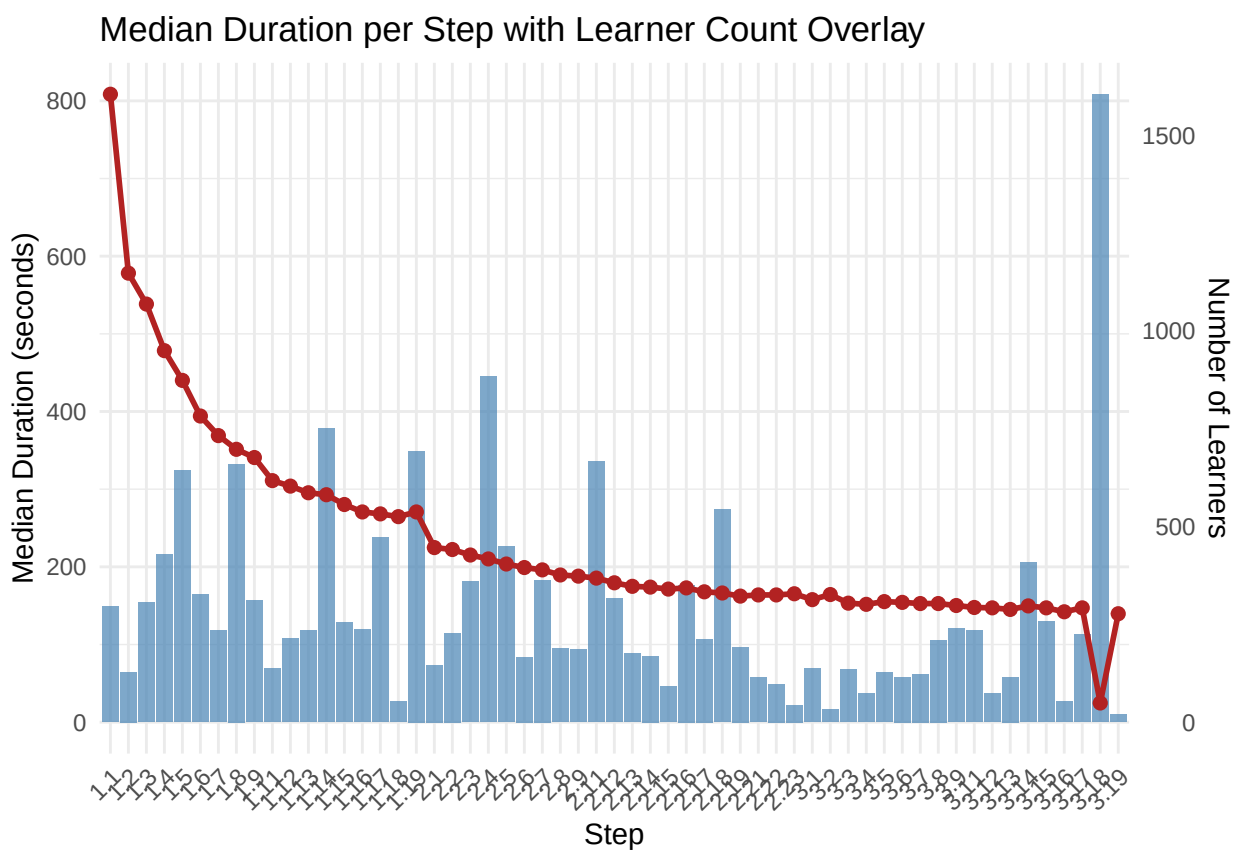



Interpreting the plot: The x-axis shows the steps in the course, probably in ascending order (1, 2, 3, . . . ). The y-axis shows the median duration learners spent on each step, measured in seconds. Each bar height corresponds to the median time learners took on that particular step. From the plot, you can see: Some steps have very low median durations (short bars), meaning learners spent less time on those steps on average. Some steps have very high median durations (tall bars), meaning learners spent a longer time on those steps. For example, there's a very tall bar at around step 3, showing a large spike in median duration, indicating that step 3 took much longer on median compared to other steps. Some steps show a cluster of short bars with very low durations, possibly indicating quick steps or skippable content. The variability between steps in terms of median duration could suggest that some steps are more complex or require more effort.

```
#time taken at eat step and the number of learners at each step
ggplot(step_summary, aes(x = step)) +
  geom_col(aes(y = median_duration_secs), fill = "steelblue", alpha = 0.7) +
  geom_line(aes(y = learner_count_scaled, group = 1),
```

```
            color = "firebrick", size = 1) +
  geom_point(aes(y = learner_count_scaled),
             color = "firebrick", size = 2) +
  scale_y_continuous(
    name = "Median Duration (seconds)",
    sec.axis = sec_axis(
      ~ . / max_duration * max_learners,
      name = "Number of Learners"
    )
  ) +
  labs(
    x = "Step",
    title = "Median Duration per Step with Learner Count Overlay"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

```
## Warning: Using ‘size‘ aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use ‘linewidth‘ instead.
## This warning is displayed once every 8 hours.
## Call ‘lifecycle::last_lifecycle_warnings()‘ to see where this warning was
## generated.
```




Interpretation: Median Duration per Step: The height of the bars varies substantially between steps. Some steps have very high median durations, such as step 3, where the median time exceeds 800 seconds (~13

minutes). This indicates that this step takes significantly longer for learners to complete than other steps. Many steps have relatively short median durations below 200 seconds (~3 minutes), suggesting quicker completion. The variation in median duration likely reflects different content types or difficulty levels across the steps. Number of Learners per Step (Red line): The red line shows a general declining trend in the number of learners as the step number increases. Early steps have the highest learner counts, close to or above 1500 learners, indicating most learners start the course and engage with initial steps. The number of learners drops steadily through the course, reflecting natural attrition or drop-off as learners progress. There are some fluctuations where learner counts temporarily stabilize or increase slightly at some steps, possibly indicating re-engagement or course structure effects. At some steps (e.g., around steps 18-19), the learner count drops sharply, suggesting significant dropout points. Relationship between Duration and Learner Count: Steps with longer median durations don't always correspond to higher or lower learner counts. For example, step 3 has a very high median duration but still retains a large number of learners. In contrast, some steps with moderate or low median duration have low learner counts, which might indicate dropout is not necessarily driven by step length alone. Overall conclusion: Learners tend to spend different amounts of time on different steps, with some steps being notably longer. Learner attrition is evident as the course progresses, with fewer learners reaching later steps. The combined visualization helps identify critical steps that take longer and steps where learner drop-off is highest, useful for course design improvements.